

SecureMailScope

Cryptographic Security Posture Assessment Report

Generated: 2026-09-08 17:38:41 UTC | Total Sessions Analyzed: 4

4	3	1	61.8/100	3	
TOTAL SESSIONS	ENCRYPTED	PLAINTEXT	AVG POSTURE SCORE	ML ANOMALIES	
CRITICAL	HIGH	MEDIUM	LOW	INFO	TOTAL FINDINGS
2	4	1	0	0	7

Protocol Breakdown: IMAP: 1 | POP3: 1 | SMTP: 2

Detailed Session Forensic Analysis

Session 6e2b7b6ee35e (SMTP) [ML ANOMALY DETECTED]		Score: 57.0/100 — HIGH RISK	
Server Endpoint	10.0.0.1:25	Client Endpoint	10.0.0.2:49100
Transport Security	Yes (Encrypted)	Plaintext Credentials	None detected
STARTTLS Negotiation	req=True upgraded=True stripped=No	Auth Command	None
Packets & Flow	6 packets (C→S: 108 B, S→C: 101 B)	ML Evaluation	Score: 0.53 Anomaly: YES

TLS Cryptographic Parameters

TLS Version	TLSv1.2	Negotiated Cipher	TLS_RSA_WITH_RC4_128_SHA
Cipher Hex Code	0x0005	Server Name (SNI)	None
Key Exchange / Group	RSA (static) / N/A	Signature Algorithm	N/A
JA3 Fingerprint	N/A	Offered Ciphers Count	0 suites
JA4 Client FP	N/A	JA4S Server FP	t120000_0005_000000000000

Security Findings (2)

[CRITICAL] tls.cipher	Weak cipher suite in use [CWE-327] [CVE-2015-2808] TLS_RSA_WITH_RC4_128_SHA uses RC4: RC4 cipher is broken for all uses; disable it. <i>Recommendation: Remove weak cipher suites from the server's TLS configuration.</i>
[HIGH] tls.keyexchange	No forward secrecy (static key exchange) [CWE-327] [CVE-2015-4000] The session used RSA (static) key exchange, which does not provide forward secrecy. A compromised server key could decrypt past traffic. <i>Recommendation: Enable ephemeral key exchange (ECDHE/DHE) with forward secrecy.</i>

Session 61e544c81dac (IMAP) [ML ANOMALY DETECTED]		Score: 55.4/100 — HIGH RISK	
Server Endpoint	10.0.0.1:143	Client Endpoint	10.0.0.3:49200
Transport Security	Yes (Encrypted)	Plaintext Credentials	None detected
STARTTLS Negotiation	req=False upgraded=False stripped=No	Auth Command	None
Packets & Flow	3 packets (C→S: 52 B, S→C: 78 B)	ML Evaluation	Score: 0.58 Anomaly: YES

TLS Cryptographic Parameters

TLS Version	TLSv1.0	Negotiated Cipher	TLS_RSA_WITH_AES_256_CBC_SHA
Cipher Hex Code	0x0035	Server Name (SNI)	None
Key Exchange / Group	RSA (static) / N/A	Signature Algorithm	N/A
JA3 Fingerprint	87e6f9d8ed4c1f40e545b4804251ffbb	Offered Ciphers Count	1 suites
JA4 Client FP	t10i010000_692296a295db_000000000000	JA4S Server FP	t100000_0035_000000000000

Security Findings (3)

[HIGH] tls.version	Obsolete TLS 1.0 negotiated [CWE-326] [CVE-2011-3389] TLS 1.0 is deprecated and vulnerable to BEAST and RC4-based attacks. <i>Recommendation: Disable TLS 1.0 on the server; require TLS 1.2+.</i>
[MEDIUM] tls.cipher	Non-AEAD CBC cipher negotiated [CWE-327] [CVE-2016-2183] TLS_RSA_WITH_AES_256_CBC_SHA uses CBC mode without authenticated encryption. <i>Recommendation: Prefer AEAD suites such as AES-GCM or ChaCha20-Poly1305.</i>
[HIGH] tls.keyexchange	No forward secrecy (static key exchange) [CWE-327] [CVE-2015-4000] The session used RSA (static) key exchange, which does not provide forward secrecy. A compromised server key could decrypt past traffic. <i>Recommendation: Enable ephemeral key exchange (ECDHE/DHE) with forward secrecy.</i>

Session 185c3888382b (POP3)

Score: 88.8/100 — **MEDIUM RISK**

Server Endpoint	10.0.0.1:110	Client Endpoint	10.0.0.4:49300
Transport Security	Yes (Encrypted)	Plaintext Credentials	None detected
STARTTLS Negotiation	req=False upgraded=False stripped=No	Auth Command	None
Packets & Flow	3 packets (C→S: 105 B, S→C: 78 B)	ML Evaluation	Score: 0.35 Anomaly: No

TLS Cryptographic Parameters

TLS Version	TLSv1.3	Negotiated Cipher	TLS_AES_128_GCM_SHA256
Cipher Hex Code	0x1301	Server Name (SNI)	None
Key Exchange / Group	ECDHE / x25519	Signature Algorithm	N/A
JA3 Fingerprint	c0fa996cd90cf490c422cc5e3291cddc	Offered Ciphers Count	2 suites
JA4 Client FP	t13i020200_62ed6f6ca7ad_efa03c98e060	JA4S Server FP	t130100_1301_d5d54e22bb3e

Security Findings (0)

No vulnerabilities or security posture findings identified for this session.

Session 0d90696c884a (SMTP) [ML ANOMALY DETECTED]

Score: 45.8/100 — **CRITICAL RISK**

Server Endpoint	10.0.0.1:25	Client Endpoint	10.0.0.5:49400
Transport Security	No (Plaintext)	Plaintext Credentials	DETECTED (Plaintext Leak)
STARTTLS Negotiation	req=False upgraded=False stripped=No	Auth Command	AUTH LOGIN
Packets & Flow	3 packets (C→S: 31 B, S→C: 28 B)	ML Evaluation	Score: 0.88 Anomaly: YES

Security Findings (2)

[HIGH] starttls	No encryption (STARTTLS not used) [CWE-319] The email session exchanged data in plaintext without STARTTLS. <i>Recommendation: Require STARTTLS and reject plaintext sessions.</i>
[CRITICAL] starttls	Credentials sent in plaintext [CWE-319] Authentication credentials (AUTH LOGIN) were observed before the TLS upgrade. <i>Recommendation: Never send credentials before the TLS handshake; enforce STARTTLS-before-AUTH.</i>

Regulatory Compliance Matrix

Automated compliance evaluation across industry cybersecurity and data protection standards.

Cybersecurity Framework	Overall Verdict	Passed	Failed	N/A Controls
PCI-DSS 4.0	NON-COMPLIANT	15 passed	9 failed	20 N/A
NIST 800-52r2	NON-COMPLIANT	15 passed	9 failed	20 N/A
HIPAA	NON-COMPLIANT	15 passed	9 failed	20 N/A

Detailed Control-by-Control Audit Summary

ID	Security Control Requirement	PCI-DSS 4.0	NIST 800-52r2	HIPAA
C01	TLS version >= 1.2	✗ FAIL Req 2.2.7, 4.2.1	✗ FAIL §3.1	✗ FAIL §164.312(e)(1)
C02	No weak ciphers (RC4/DES/3DES/EXPORT/NULL)	✗ FAIL Req 2.2.7, 4.2.1	✗ FAIL §3.3.1	✗ FAIL §164.312(e)(2)(ii)
C03	Forward secrecy (ECDHE/DHE)	✗ FAIL Req 4.2.1	✗ FAIL §3.3.1	✗ FAIL §164.312(e)(2)(ii)
C04	AEAD cipher mode (GCM/ChaCha20)	✗ FAIL Req 4.2.1	✗ FAIL §3.3.1	✗ FAIL §164.312(e)(2)(ii)
C05	Valid, trusted certificate	— N/A Req 4.2.1	— N/A §3.4	— N/A §164.312(e)(1)
C06	Certificate not expired	— N/A Req 4.2.1	— N/A §3.4	— N/A §164.312(e)(1)
C07	Strong public key (RSA >= 2048 / EC >= 256)	— N/A Req 2.2.7	— N/A §3.2	— N/A §164.312(e)(2)(ii)
C08	No SHA-1 certificate signatures	— N/A Req 4.2.1	— N/A §3.4.3	— N/A §164.312(e)(2)(ii)
C09	Encryption required (not plaintext)	✗ FAIL Req 4.2.1	✗ FAIL §3.1	✗ FAIL §164.312(e)(1)
C10	No plaintext credentials	✗ FAIL Req 8.3.2	✗ FAIL §3.5	✗ FAIL §164.312(d)
C11	No STARTTLS stripping	✓ PASS Req 4.2.1	✓ PASS §3.1	✓ PASS §164.312(e)(1)